

ActInf GuestStream #033.1 ~ "A taxonomy of surprise definitions", Alireza Modirshanechi

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hello and welcome everyone it's active guest stream 33.1 January 17 2023 we are here with Ali and alireza modir shanichi we're going to hear about the taxonomy of surprise definitions there will first be a presentation followed by a discussion so Ali Reza thank you for joining please take it away with the presentation we'll be looking forward to this and the discussion thank you very much Daniel and Ali for inviting me um so as you said I'm going to talk about a taxonomy of surprise definitions uh which is um basically all the materials or most of the materials that we publish in this recent paper in the Journal of mathematical psychology uh that is a joint work with johanni Brea and Wolfram gerstner and lufranker is my PhD supervisor so talking about surprise let's start with a simple top experiment um let's say that you want to plan something for your weekend and to plan something for the weekend we need information like how the weather is going to be or we can plan something and or some cues like the spring that are informative about the weather of the weekend or that uh the weather forecast says is going to be sunny given these queues we can make some predictions that oh weather forecast said it's sunny and it's spring so most likely it's going to be sunny let's say that is Friday evening you plan that for example go for a bike you go to sleep wake up the day after uh open the window and it's snowing and seeing this note given that you expected the sunny weather this mismatch is what's been uh natural language would agree on that it's a feeling of surprise you know we can easily talk about this moment say that yeah I was surprised because I had planned to go for a bike but then in scientific Community we can ask this question that for these moments that you see a snowy weather when opening the window what does really happen in our brain and uh lots of literature in a long lasting debate in neuroscience and psychology about different roles of surprise in the brain and there's this slew of different uh what I call surprise related signal things like prediction error free energy surprisal or Bayesian surprise and people say that these things are related to learning play important role in model building or predictive coding some people say that they're quite

important for exploration for example surprising even attract attention or drive curiosity uh there are also literature and memory that surprising events um segments or continuous stream of observation so they help us to segment our memory and of course there are lots of work and physiological signals of uh surprise signals for example when there is something surprising your pupil dilates or EG signal has a peak and so on so forth the thing is that if I enable look at this picture and from far away it seems like that surprises everywhere in the brain but then the puzzle is that it's really the surprise in the sentence surprise martial synaptic elasticity same as the one in the sentence surprise Drive curiosity or not that's basically the motivation of the main motivation of our work that uh different studies in neuroscience and psychology use similar words but to refer to different aspects of the moment of surprise and I would call this art piece the moment of surprise and uh basically the thing is that different studies different experiment different paper they all talk about moments like these situations like this that we can all agree on to call it a surprising moment but they talk about different aspects of it the way that some we summarize it or put it in verb in words in the abstract of the paper was there is no consensus on the definition of surprise when I posted the paper online on Twitter the linguist margin has to laugh wrote in reply that it's unsurprising that there is no consensus on the definition of surprise because it's not a technical term everyday words are usually somewhat awake and uniform definitions are desirable only for technical terms of science and I cannot agree more with that and I want to emphasize that today I'm not going to talk about the meaning of surprise when we talk over a coffee break it's in the context of neuroscience and psychology and when we talk about these moments and the role of these surprise related signals in the brain so people are going to details and math and all the framework um I want to present the paper in a nutshell that what is the Absurd of the paper what we do is that we take many papers from the field and we identify several definition of surprise at least 10 of the definitions here in the paper that are 18 with all the details and these definitions include things like ansar reward prediction error confidence corrected surprise Bayesian surprise positive surprise and many things that people really use in different contexts then we ask this question that what are their similarity and differences between these different surprise definition and we have three main results that is uh analyzing these different definitions from different aspects well first main result is a technical classification so to talk about the technical classification I would like to introduce a short notation we call observation y we call Q 's X things that are going to be informative about the observation and then predictions can be seen as either a probability distribution over the observation given the cues or one estimate of the next

observation and looking at this uh surprise definition we could um identify a category of these definitions that what we call observation mismatch to price definition they only depend on the next estimate so they don't care that okay what does the probability of weather being sunny or rainy or snowy they say that I predict that the weather is going to be sunny and now it's not sunny and what is the difference between being sunny and being snowy and that's the surprise there is another group that we call probabilistic mismatch they actually look at this distribution and say that oh uh it's true that I didn't expect the snowy weather but it was not too unlikely so it may be not as surprising as I would expect it so they work with the with a distribution instead of a single point estimate or predictor um actually this uh conditional distribution is a marginalization of something bigger because there is actually some uncertainty about what how accurate for example weather forecast is about the weather of the next day and that what we would call I would put in this parameter Θ that are underlying rules of the environment or hypothesis the reliability of the weather forecast and we don't have access to the true value of it so we make it believe and this P_i of θ is our belief it's a distribution over these possibilities and actually the marginal distribution of next observation given the queue is this marginal probability and the third group of the surprise definitions people defining these definitions uh argue that actually what we care about is this belief that for example how much we learn how much we change this belief uh by observing a new new observation and those are the definitions that we call belief mismatched surprise definitions so this was a technical classification we put different definitions in three different groups and say that there is some uh difference in how they depend on uh the belief of subject then we go to a conceptual labeling and that's what we call the taxonomy and we talk about what is the concept or what is the conceptual arguments behind the definition of each of each of these surprise definition one group uh that's what we call prediction surprise so they uh they care about how accurate or inaccurate the prediction is how expected or unexpected the next observation is the more accurate the less surprising the observation then there's another category that we call change Point detection surprise they don't only care about the prediction they always compare prediction with a baseline prediction if something is unexpected but that thing is unexpected under any hypothesis then they claim that is not surprising because it would be just kind of uh an outlier so I should not consider it as surprising and these are important to detect change points um because what they care about is to see if there is another hypothesis under reach the current observation would be more likely for example in this case of weather forecast during covet time there was this argument that there is this paper I'm citing here that because there is

much less flight uh going on we had way less data and sensors so weather forecast suddenly stopped being accurate and actually if you were checking weather forecasts you could see uh without even knowing the news that uh it gets less and less accurate as time pass on and this kind of change Point detection surprise definitions are somehow optimal I show later on that they're optimal for detecting these kind of unobserved changes uh the third group or the third class are what we call Information Gain surprise so no matter if there is a change or not uh this new observation made me learn something more about those underlying rules of the environment so uh which Media or which weather forecast is more reliable than the others every day that I see the weather I would use that to update my belief about different weather forecasts and the last category is no matter which of these ones I care about what what is the concept behind the what is the conceptual argument behind the definition the argument is that if I'm more confident if I'm more confident about my prediction I should feel more surprised so the argument is that no matter what is the definition I should have an explicit term for confidence in my definition so having this technical classification and this conceptual labeling we can arrange this list of surprise definition in this different boxes and have some structure and as you can see some of these boxes are empty and I would like to say you know we have these mathematical definitions on the side but their justification come from the relation to experience uh to uh experiments and uh what they tell us about the brain so it can be the case that in future we see that none of these definitions are helpful or we see in some situation that we need another definitions and this uh figure here tell us that okay there are actually a places that we can come up with new definitions in new categories but there are also restriction for example talking about Information Gain if we want to talk about Information Gain we should always talk about the belief mismatch we cannot talk about only at the level of observation or marginal distribution because we are talking about learning so we should talk about these very high level uh hidden variables or when we talk about confidence we inevitably we need to talk about probability distribution and we cannot talk about Point estimates but uh still we can talk about these three different uh empty boxes here and or last main result is although these different definitions are in different boxes but there are some links between them so we found conditions under which these definitions are indistinguishable so there are some situation that they're wanted on mapping uh there is one-to-one mapping between some of these definitions we cannot really experimentally distinguish them so um that was a summary of our work and now I'm going to go uh to the mathematical framework and go one for one through all of these definitions and they're linked to each other and please let me know if there is any um

clarification question so for mathematical framework Let's uh look at one very traditional experiment experiments for studying surprise in the field of Neuroscience it's what what's called volatile audible task uh in a typical output task at each time point we have two possibility for uh the stimulus it's either for example here a blue disc or a red square and for a while the blue disc is much less likely to appear on the screen than the other one and the reasoning behind the design of this experiment is that the participants who is watching this sequence on the screen one after another expect to see a the red screen more often than the blue disc and at the moment that it sees uh he or she sees the blue disc it's uh it's surprising and that would give them a feeling of surprise and a question that one can ask is that can we really quantify how surprised a participant feels at time t equal to four so if we know that this is the sequence until time t equal to 4 can be Quantified the amount of surprise or not and to talk about the feeling of participant about surprise of a surprise of an observation we should make some assumption about how participants perceive this sequence and we take it very common Assumption of the field that participants perceive their sensory observation as probabilistic outcomes of a generative model with hidden variables so at each time each of these observation come from some distribution with some hidden variable for example the way that participants we assume in a way that the way that participants think about the sequence of observation is that at each time with some probability uh one of these uh stimuli appear on the screen and over time given the sequence they try to find this probability uh to formally Define this we consider a generative model let's say that at time T we show the observation what appear on the screen by Y_T and there is this hidden variable which we call environment parameters which is for example the probability of blue disk appearing on the screen and now this link for people who are familiar with and going to make a Bayesian Network and this link shows that the environment parameter determine the distribution over the observation and I can repeat the same thing for time t plus one something that I didn't talk about in these volatile Oddball task is that for a while we had this uh one image after another with some fixed probability but once in a while we have a hidden change point which changed the probability of blue disk and red square so previously bluedisk was rare and after that the other one can be rare or frequent so the environment parameter over time can change so we defined this parameter uh this random variable change Point indicator that at time t plus one if it is one it means that there is a change in the environment there is a new parameter and if it is zero it basically means that there is no change the environment is stable and Θ_{t+1} is a copy paste of Θ_t So this link just take that a t and take it to θ_{t+1} um so far

the way that I present this in uh this task uh was that different observations are independent of each other so at each time one of these stimuli appear on the screen randomly um this very beautiful paperless manual and uh colleague in 2016 in Pillars computational biology they argued it's very nice they argue that even if the sequence of observation is randomly uh designed so this observation are independent of each other people when they are observing this sequence assume that there's some dependency so they what they uh what they estimate over time is not really probability of observing the blue disc or Red Square they are estimating the probability of observing blue disk after Red Square or Red Square after Red Square so what they assume is that there's a link between y_t and Y_{t+1} so so far I have a model of this uh Oddball task that says that each observation depends on the environment parameter and the previous observation and there is a dynamic for this environment parameter I do a bit of trick just Define this dummy variable which I call Q and for the Oddball task I just copy paste the previous observation in the queue but it uh the reason that we use Q is to have a one abstract notion of what is um what has predictive power for the for predicting next observation and for the for the task with the action action can go to Q for the example of the weather forecast weather forecast can go to the queue and all these things and um now having this Motif I can repeat it and make the whole sequence and given this generative model given this mathematical model of the experiment I can now mathematically Define the surprise of observing variety plus one so with this assumption we can uh we can propose some quantification of surprise of one stimuli and uh what is force mentioning is that this generative model here is generalization of many other models in the field which means that many of the experiments already done to study surprise can be modeled by this generative model and a results hold for that okay um before going forward this generative model also account for these uh moment of surprise can model that that the Q was X is here the observation was the weather of next day is here and Θ was environment's parameter was how reliable for example the weather forecast is about the next observation and the change point is whether something happened in the environment for example the pandemic and decreasing the number of planes and now I'm going to make this definition more formal and discuss the Dynamics of each of these different levels of in the generative model so I start with the observations and we assume this is the probability of at time $t+1$ observing y condition on Q being X and parameter being Θ and basically this equation here means that this probability is time independent so as long as I know the Q and the parameter it doesn't matter that if it's time Step 1 or time step t in all the time the same distribution which we uh show by pfy given X so

basically any change in the environment goes to the parameter and the distribution is fixed um for the Q variables we don't put any constraint and for example it can even be independent of anything that's object assumed which is the for example for weather forecast uh you just turn on the TV and see what's going to be the prediction for next state then we have the change points indicator and their Dynamic is at each time independently we ask whether there is a change or not and CT plus one and time t plus one is a bernardly binary sample with probability of PC which is the chain what we call change Point probability so at each time point with probability PC there is either a change or not then um this is the most important in a way part of the model that how this parameters change over time we consider prior belief Pi zero and say that at time one before the experiment starts participants have some uh assumption and some belief about what are the parameters Behind these uh generative process and uh that we show with pi zero and at time T if there is no change in the environment if c t plus 1 is equal to zero Θ t plus one is just copy paste of θ T but if there is a change in the environment we assume that Θ t plus one is sampled again independently of past from this prior belief so we have a stable environment with the same data until CT plus 1 is equal to one then we have a change in the environment and a parameters is uh resampled from these prior belief these equations that I wrote here is fully describe the joint distribution of $C1$ to c t plus one Θ 1 to T Θ t plus one and so on and so forth the whole generative model is described by these definitions but if you want to see the precise mathematical definition please look at the definition one in the before going forward I would like to show some notation that uh whenever I write something like this I shortened the vendors whenever there's no ambiguity I shortened with the notation by dropping the random variables that are uh the capital letters um then I Define what I call observers belief at time T if you remember Pi zero I Define it as initial belief and now Pi T is the Observer belief at time T that is given that I observe $y1$ to YT with q 's $X1$ to x t what is my knowledge about the current parameters of the environment which basically can be summarized in the posterior probability of θ T given the previous cues and observation which in this example is that given a sequence of days and a prediction of the weather forecast I can make a belief on how reliable the weather forecast is at time t and at the end they Define this marginal probability which is the probability of Y given X and now I dropped Θ and I put the knowledge here and integrate over all possibility that if the world is at time t how my knowledge about the world identify the relation between q and observations so this was uh the mathematical framework and given this mathematical framework now I can go to a formal investigation of different definitions and uh present or

classification in distinguishability condition and taxonomy what is surprise having this model I can ask this question that condition on the previous observations y_1 to T and Q variables X_1 to t plus one these ones and these ones how surprising is the next observation Y_{T+1} we are at time t we are going to observe Y_{T+1} and this is now a valid mathematical question in this framework to make some prediction about Y_{T+1} I need information about T and x_{t+1} that is the Q and these are the rules of the environment whatever I know about the rules of the environment is summarized in my belief in my uh in whatever knowledge this uh observation and previous cues gave me about t and uh given the belief and the queue I can get rid of I can put away throw away all the previous observation and say that I only need the belief and the Q and surprise is basically a function that gets us arguments the belief the uh the queue and the next observation and give a real value back one famous example is what people call surprisal or uh Shannon's surprise the history is so long it has been used for so long that I cannot really going to the first paper that use it but you can look at the paper of barter in 2013 for a review which basically space IQ and an observation surprise of uh this observation is equal to minus logarithm of marginal probability so the more likely this observation the less surprising it is there's also another way to define it which um because here I had the knowledge at time t one can say that there was a possibility to have a change Point here and uh one can extend this probability and say that oh there was a when I'm at time T and I'm predicting the next observation I should also consider this possibility that the Vault may change in between and I should put some weights on this prior knowledge so the next observation comes with some probability from the current Volt or from the uh prior to terminal word that is recent uh so there are two possibilities for these definitions and um what I want to emphasize is that in volatile environments we are not sure which one we should take necessarily one may say that the this definition makes more sense because it's the full probability but there are experimental evidence in this paper of Nassau and colleague in 2010 in the Journal of Neuroscience what they argue is that when people observe Y_{T+1} they update their knowledge by assuming that there might have been a change but when they predict the next observation they don't consider that there may be a change so in a way when people are thinking about the future they don't consider this possibility of a change um and in this sense uh we argue that the first definition uh the one that I call S of $sh2$ is more consistent with the experimental evidence but through the paper we take all these two versions it appears for many of the definitions and we do really this tedious job of keeping all these definitions through the whole paper but today in the talk I only present one version of each definition so

this was one of the examples I can have I can present different examples of surprise definition in the field uh with three of them I'm going to present the first uh classification that we have the first one was Channel surprise another one is the absolute error as I said before one can say that given the belief that I currently have I can make a prediction for example take the average of these uh marginal probability and then the difference between these average these that was my prediction uh and the difference between the average and the true observations in the actual observation is defined as a surprise is some sort of prediction error and there's also this other definition of very famous definition Bayesian surprise that says that I don't care that uh what is the difference between Y_T or $\hat{Y}$ and what is this marginal probability what I care about is that by observing up and observing this Y_T plus one how much I update my knowledge about the world and what is the difference d_{kl} here stands for Kullback-Leibler Divergence between the belief before observing Y_T plus one and after observing Y_T plus one um it is not written in the form of Y_T Plus One X plus one and P_i T but uh in proposing in remark 2 in the paper we show that if PC if probability of change was equal to zero so if there were no change in the environment Bayesian surprise could be written as a difference between expected Channel surprise and Shannon surprise itself and funnily enough the channel surprise here appeared with the negative sign so um it seems like that the Bayesian Superstition Channel surprise are really complementary uh definitions and uh particularly it is important to say that we cannot really get rid of uh the belief belief really appears remains here so we have marginal probabilities here but we also need the belief to evaluate Bayesian surprise for the case of a volatile environment with PC greater than zero we show that in Proposition 8 the interpretation remained the same but the equations become a bit more ugly so that's why I don't bring it here but you can check it out having these three definitions here something that's important is that piano surprise could be written in terms of the marginal probability absolute error could be written in terms of the $\hat{Y}$ hats for an estimate of an X observation but for Bayesian surprise we needed the whole belief because we had to compute this expectation and that's how we make our first technical classification which is based on the dependence on the belief that this surprise definition how it depends on belief pie here on top we have PI from PI we can extract the marginal probability and from the marginal probability we can extract one estimate the first group are the observation mismatch surprise definition that only depend on the estimate for example the absolute error that I introduced the second group are the one that only depends on the marginal probability there are one like a Shannon surprise that is minus logarithm of the marginal probability but the last group they cannot be evaluated by only

knowing these two variables by knowing this distribution and the next prediction denote the whole belief like Bayesian so going back to this uh figure that I show in the introduction so far we have seen two definitions here in this column two definitions here in this column and two definitions here in this column the first two absolute error and shine on surprise were more about the prediction so this one was saying that what is the mismatch between the prediction and observation and this one was saying that How likely or unlikely observation was while the Bayesian surprise was saying that how much information I gained upon this observation okay this was a technical classification I can go forward now and discuss the indistinguishable conditions before going to indistinguishable condition I want to introduce another definition of surprise there is this uh opinion this idea in the field that surprising events increased learning rates and there are some theory for it and there's some experiment showing that it actually happened so then there is something surprising the learning rate increases but there is this um logical question that is increasing learning rate always good for making better predictions or making better knowledge of the environment or not because uh sometimes something surprise something can be unlikely but if it is just outlier maybe I should not change my learning rate I should just ignore this observation what we find out um is that there is a is necessary to have a comparison between the current belief and the prior belief because basically the moment that you want to increase your learning rate upon surprising events is when there is a change in the environment and you want to say that I want to get rid of the old and this base Factor surprise we proposed it in a paper in neural computation in 2021 and it always compared the probability of observation under the prior belief with under the current belief it's some sort of hypothesis testing at each point saying that oh whether there was a change or not so if this um the nominator the probability under the prior belief if we ignore that this is basically a decreasing function of the probability under the current belief very similar to the channel surprise so there's the less likely an event the more surprising it is but if that event is also unlikely under the prior belief then it may not be surprising and um the reasoning become more valid when we show in our proposition one in the paper that actually this definition is optimal to modulate learning in proposition one we show that after observing Y_T plus one the new belief can be written in this term which is a trade-off between integrating new observation into the previous belief so take the previous belief take the new observation combine the two make a new or throw away the old belief say that oh there was probably a change in the environment I should reset from scr and start from scratch and uh put the new observation together with the prior belief and make a reset belief and this trade-off is controlled by

something that we call adaptation rates here shown by $\gamma_t + 1$ which turned out to be an increasing function of Base Factor surprise so the higher base factors surprise the higher adaptation rate and the higher adaptation rate the more weight on the reset basically saying that the surprise uh and this rule really comes out of the Bayesian inference is a normative rule and uh basically saying that base factors surprise uh naturally emerge or at least we Define it in a way to have uh this decomposition of the belief in this paper we propose also a variational method that turned out to be have a very simple update rule for distribution in exponential family which we call a variational surprise minimization learning and I would like to mention here that recently it got some applications in the active inference community and people used it for model building so I introduced the base Factor surprise as a new base uh surprise definition let's go and just overview the definitions that so far I showed the first one was absolute error surprise was in the family of observation mismatch and it was it care about how good my prediction is then there was Channel surprise saying How likely or an unlikely an observation is it was again caring about prediction but it was in the probabilistic mismatch the base Factor surprise that I just showed isn't the family of change Point detection because as we saw it care about whether there was a change point or not but it is still in the probabilistic mismatch to price definition [Music] and another definition that I showed was uh Bayesian surprise which was the difference between uh prior previous belief and the new belief and is in the belief mismatch to price definitions um although they are in different columns uh in different boxes one can ask this question that is there any link between Shannon's surprise uh here and the base Factor surprise here because at least there's one term that is repeated in both definitions that's what we show uh in the preposition two of the paper that Shannon's the the base Factor surprise can be written as function of these interesting things here that are not Channel surprise but very interestingly related to the channel surprise there are difference in Shannon's surprise so there are different between the Shannon surprise under the current belief and under the prior belief base Factor surprise can be written as a function of difference in Channel surprise so one may say that a base Factor surprise has a interpretation like a relative surprise and in that sense it always compared the prior belief with the current belief and that's why it's good for learning as a consequence in corollary one you can argue that optimal surprise modulation of learning can also be written in terms of these differences instead of the base factors to price but the most fun thing is that in corollary 2 we can we show that if the marginal probability under the prior belief is uniform so if basically it means that if this term is constant then there are strictly increasing mappings between base Factor surprise and two

definition of Shannon and strictly increasing mappings basically means that these are one-to-one uh function of each other so they're invertible and if I have one of them I can find the other two so in a way they're indistinguishable and that's one of the big results of our paper that's the uh I identify the conditions under which different definitions are indistinguishable so here in this figure you see different definitions they are connected to each other with lines each line corresponds to one of these conditions that if this condition is satisfied then these two definitions are indistinguishable and uh C2 here means that this condition is proved under corollary two in the paper why this uh indistinguishability conditions are important um the first thing is for experiment design for example let's take the um Oddball test that I introduced before usually an oddball task it is assumed that uh marginal distribution under the prior belief is flat because there's no preference for the uh at the beginning there is no preference between the blue disk and the Red Square as a result basically means that in this kind of experiment all these five definitions that I have here are indistinguishable so I cannot do the Oddball task and expect to see a difference between these five definitions in addition some people argue that even the prior belief is flat and if is that the case then these two definitions are even more because Oddball task is a categorical task it is one of the special cases that we identify which basically means that seven of these definitions are indistinguishable so it's quite important that if we want to make experiment to distinguish some of these uh surprise definitions to study different definitions of surprise we should be careful to not have experiment where these are indistinguishable and what we do is that we take a series of papers in the field that have experiments where they study different definitions of surprise and we identified which of these conditions are satisfied or not in these papers the second thing that these industry increasability condition can be important for is for computational modeling there is this page this new preprint from orlab and Via archive they're um the authors show that they design a spiking neural network that try to learn in a volatile environment and modulate its learning rate with the notion of in the setting that they analyze we can show that these three lines the condition corresponding to these three lines are satisfied so basically these definitions of surprise are equivalent what is uh why this is interesting for us because this one is optimal for learning is the base Factor surprise that I showed before while this one is pretty easy to implement while base Factor surprise was a function of probabilities well this one is the observation space and that's actually the fun thing that what they do here is they make a spiking Network that computes something related to this observation mismatch surprise definitions while there are some uh optimal justification for why these um surprise definition

can be helpful for so putting together so far I proposed the I presented the technical classification that we proposed and indistinguishability condition that we found out and now I'm going to discuss the I had 40 minutes so I still have time right I can so for your taxonomy I need to introduce another definition I promise is the last definition that I uh introduced today it's um the notion of confidence correction I want to introduce that and there is this idea that higher confidence higher confidence must lead to higher feeling of surprise so the more confident you are about your predictions the more surprise you should feel energy at all uh in this paper proposed this definition uh it may look a bit strange at the moment I would demystify it in a bit it's the difference between the current belief and after observing this how much uh what would be my belief if I would cons if I would throw out all the previous observation and make a new belief so it's a next belief but with the assumption that there was a reset just for a comparison between the confidence corrected surprise and Bayesian surprise in the base and surprise I had a difference between the current belief and the next actual updated belief but for the confidence corrected surprise is the current belief with the reset updated belief uh the reason that is interesting is that in proposition 9 of the paper we show that this definition can be written in this way where the first name is different in Shannon surprise the second term is kind of difference is based on surprise and the third term is a notion of confidence so this one looks like a relative surprise as I argued before it's good for uh modulation of learning and that was actually the original plan of Heritage at all to find the surprise definition that is good for surprise modulation and the last term is a notion of confidence and in these pre-print online we show in some situation uh it seems more intuitive to have some confidence correction in the definition but I don't want at the moment to um enter that discussion I just want to introduce that there are this class of definitions with a confidence correction an overview of all definitions so far the absolute error but the difference between observation and prediction in this class Shannon's surprise how unlikely something is base Factor surprise uh how unlikely something is compared to a baseline or prior belief the Asian surprise how um how much I gain information upon observing something and confidence corrected surprise I should include the explicit notion of confidence having these five definitions now we saw one definition per row and one definition or at least one definition per column now I can introduce our taxonomy on the on the screen now you see in a seemingly arbitrary order these definitions that I had here and we propose to label these different definitions based on the quantity they measure this group that's what we call a prediction surprise they care about how good your prediction is about the next this group

where information gains surprise they measure how much you learn based on a new observation and these groups for change Point detection surprise that were some sort of relative measures of surprise and we had this uh overlapping labeling that no matter where there is a prediction or change Point detection we can make the effect stronger or we can correct the effects by adding an explicit term for uh confidence one can I ask that okay so far I talk about vendor indistinguishable and put them in different category one can ask how different are these definition in practice because if I um pick a random Oddball task from the field look at the data all these definitions look quite correlated and uh one may argue that okay does it really matter to have different and um it is actually possible to design experiments here I'm showing one example I'm not going to discuss that what is the experiment to be presented in this preprints on the x-axis you see time on the y-axis average surprise and four definition from these four different categories and here is a change point in the environment and we see after change point they have a completely different Behavior one goes down one has Spike one goes down and go up again and down again so there are possibilities to make Theory or even experiments that different definitions of surprise can have very different behavior and different predictions another example is um in this preprint the focus and the role of different surprise definitions and uh curiosity or even exploration that how differently they can drive exploration and um there are also many other works and that are not from our lab and they work and they focus on the physiological signatures of different definition of surprise particularly this one is in Pillars computational biology I found it quite impressive they found signatures in EEG for all three groups of prediction surprise confidence correction surprise and Information Gain surprise so a fixed tax a fixed experiment looking at the EEG they could see that all these three have their own correlates or signatures in the EG signal so I introduced the mathematical framework and formally discussed many definitions of surprise we had this technical classification and distinguishability conditions and the taxonomy take on messages one surprise in different experiments refer to different these different definitions can be classified based on on their dependence on the belief on the subject's belief and the quantity they measure which was the taxonomy or the labeling we proposed and the third message under specific condition some of these definitions are indistinguishable and I argued that it is useful for experiment design and also useful for computational modeling so putting everything together back to this figure that I showed at the beginning of the presentation our proposition is to replace or at least order these different signals or in a structured way that we can now uh if you put it in this structured way let me rephrase it in this way that then we can talk about

that oh for example for exploration reaches these different domains is more relevant is uh the belief is my surprise definitions that are more relevant to acceleration or the observation is my surprise definition so we are not talking about only specific definitions but more like categories um in addition to this uh surprise related signal or these surprise definitions we argue that there's also a need for uh novelty signals or novelty we argue in this paper that novelty and surprise are fundamentally different so everything that I said today would not uh stand for novelty so I would like to thank my um collaborators for this joint work or very cool lab in lausanne and you for your attention thank you very much and you all right just getting things back awesome I will restore our videos well thanks for that awesome talk if anyone has a uh question in the uh live chat please feel free otherwise Ali if you'd like you can re-enable your video and feel free to give a first comment or I'm happy to give a comment but you can go first uh yeah sure but for some reason I can't activate my camera here uh let me check it again okay uh so thanks a lot uh first of all I congratulate you for writing and your of course your co-authors for writing such a fascinating paper thank you I truly enjoyed it uh well I have some um comments if I may and um also I have uh maybe a couple of questions your work actually these kinds of classification work or the taxonomy is in my opinion is really a significant contribution to the whole area of psychological or especially in cognitive Sciences because as Muhammad Ali khalidi's recent book The cognitive ontology claims this kind of taxonomic approaches has been long debated among the neuroscientists psychologists and cognitive scientists especially uh about the difference between cognitive kinds and natural kinds and whether or how they map unto each other because as you also mentioned we have a folk psychological notion of surprise or uh let's say cognitive uh kind of surprise but we're not always clear about how to translate that into precise mathematical terms so I remember in active inference book by par pazulo and Kristen there's about a paragraph or so distinguishing between the folk psychological notion of surprise and the statistical surprise or surprisal functions corresponding to various distributions so basically as I see it your paper is a kind of expansion on that single paragraph to consolidate all the different conceptions of surprises into a whole unified framework we were talking the other day about the importance and significance of the recent move toward applied category Theory as a kind of as I like to call it the conceptual housekeeping or better still a kind of cartography of the knowledge trained so this this kind of research in my opinion can be seen in congruence with this recent move toward as Toby Sinclair Smith would call well to do a kind of well-typed science to be precise and clear about our definitions uh and uh I also wanted to uh kind of getting back to that uh

distinction between the cognitive kinds and natural kinds um there are different taxonomic approaches in cognitive science addressing these problems uh but uh as I understand it uh your work here is more in line with the revisionism approach to do these taxonomic uh these classification approaches which says that which which basically says the current cognitive taxonomy is not precise enough to adequately address uh the problem of uh distinguishing between cognitive kinds and natural kinds so we need a more precise formalism to do that reminds me of uh Dalton's active advil's notion of blanket index introduced in his paper a week a mark of blankets because before before the publication of that paper there was much confusion and ambiguities about the exact mathematical definition of the markup blanket or to put it in other words to make it more General and more precise what do we mean exactly by Mark of blanket so my question is as I see it uh so far you have a somehow mapped these kinds of different surprises in a kind of discrete space as you showed in one of your tables so do you see it as a feasible move forward or let's say a feasible move toward more General conception of this framework by conceiving this unified framework as as a two-dimensional continuous or n dimensional continuous manifold of possible definitions of surprises and maybe even Define an index to navigate this continuous manifold space because in that case obviously you could the problem possibly avoid um those kind of blank blank spaces you have on your tables and maybe even we can gain a kind of a much deeper insight into how exactly those different kinds of surprises relate to each other in a much more continuous sense rather than the discrete stance so do you think this kind of move even feasible from your viewpoint now that's a very good point uh thanks a lot for the for all the remarks I enjoyed it very much the way that I Define surprise in a general sense was this function or functional that get the Q observation and the belief and map it to a real value number so all the types of functional in this space can be seen as a measure of surprise so the whole Space is quite big and uh continuous uh so in principle yes of course we can do that but the space is too big actually we need to put some constraint on it so I think for future work if one want to go to that direction uh one should say that what are the properties from a surprise definition that I would like to have so put some constraint on this space and then in some sense uh parameterize this this space to have some interpretation because at the moment I'm saying that okay let's look at this part of this space that these definitions were there people are already talking about these definitions and I would call these definitions for example Information Gain surprise definitions so maybe one can really uh formalize all the definitions that it is possible to call because okay I only talk about these two particular definitions with kale Divergence if you

replace scale Divergence with any measure of uh distance on the space of uh probabilities we end up with another information in Surprise definition so I think it is feasible in principle but I think it's a bit of a tedious job and uh whether how useful it's going to be is not clear for me because we already have many definitions and as I try to in a way say what justify definition is at the end of the day how useful it is to explain experimental evidence and help us think about cognition Justified true belief distribution thank you irreza want to add a general comment and then bring us back to active inference so in exploring the active inference ontology we've often found this tension where a given term would have one definition very folk psychological very everyday belief surprise observation sense action these are some of the most common words in any language and then your work started to dissect or characterize due taxonomics on that Iceberg and show that the pointer of a given word surprise in this case can actually host uh diversity of technical definitions with some subtle contradictions or compatibilities and areas of indistinguishability so I think it really um raises our attention to the Horizon and shows that for a lot of terms that people again use in a day-to-day context there may be multiple definitions and there's many implications like there isn't a single way to model action or belief or surprise there are different flavors and even within a flavor there's going to be different ways to deploy it in a given model or script or modeling context so I think the formal coherences that you're providing are going to go a long way to ensuring that given that kind of pluralism in modeling and statistics which has always been the case that we do have connections that are robust to enable comparison like you did with different empirical setups whether the experimenters understood sh1 versus sh2 we can um post hoc essentially as peer review annotate or augment the definitions re-partition different ways that they analyzed data and so I I think that really opens up a lot of directions something that you had on several slides but didn't really go into that much was F star so could you add a little bit on free energy what is free energy what's free about it what's energetic about it why do we use it to help us engage in perception cognition action in the free energy principle and active inference thanks a lot for the question um the way that I see free energy in the context of what I discuss about is as long as I talk about exact Bayesian inference and really have my exact update rule for the uh for the belief going from the previous observation and previous belief to the exact next belief uh free energy doesn't play much of a role free energy plays a role as soon as I want to talk about approximate inference and go to the variational setting and in that case basically free energy defines um some loss between my approximation and the exact true belief and one can show that this is the upper Bound for the Shannon surprise that they Define and the

difference between the the free energy and these uh final surprise is how good my approximation is and that's why if I do exact inference the minimize for energy is equal to the channel surprise so in that sense I cannot really distinguish minimize free energy from Channel surprise but if I work in the uh approximate inference setting then minimize free energy is the best approximation of Shannon surprise that I can get by like this in that framework yeah awesome just to kind of restate that because it's actually one of the fundamental aspects of the approach taken in active inference if the true state of the world is a gaussian and we're doing exact inference with the gaussian prior we will seek to minimize our Shannon's we could if the world is not truly a gaussian but it's some other unstated or unstatable distribution but we choose to do variational approximate Bayesian inference with a gaussian prior then free energy tracks us doing strictly as well as we could given our choice of Prior family and so it operationalizes with approximate Bayesian computational techniques these formal definitions which may or may not be tractable in large State spaces or with large amounts of data so it's a super important point and it's almost like there's that Iceberg of surprise definitions and then free ports us into the approximate which enables us to get some tractability in settings where inexact Bayesian approach is not plausible okay two things first is even more than that so uh even no matter what is world I may assume that world is a as a quotient distribution let's say but there's some dependency and I still even though I'm aware of this dependency I can assume that there's no dependency and do that in my uh inference so when I think about the world it is actually true I believe that there are dependency but they still do approximate inference so I think when we do variational inference we may not necessarily care that what is the true distribution we assume the distribution has this form the yet we want to approximate it with something simpler uh this is the first part and the second part all the definitions of surprise that I talked about can be applied to this approximate uh setting as well so I can use the belief approximate belief that I found with variational uh inference and just imported in any of the definition that I had and come up with some definitions of surprise I think what I want to emphasize on is that variational free energy and minimize free energy is one of this particular one but all of them are still applicable in the approximate variational setting thank you for the correction like you mentioned variational free energy and there is expected free energy about future but also people have proposed variants like the free energy of the expected future and so it speaks to categories of functionals that do different computations and there isn't just one specific functional form they might highlight different aspects as your plot of surprisal through time let's just say it was some

stimuli and I think that leads to um a next question which is if we're interested in some biological phenomena how does this work help shape our experimental design statistical power analysis discussion section like how do the thousands of papers surely using surprise in a qualitative or a quantitative way how do we translate this work to increase the rigor and interoperability of those studies that's a very good question uh and that's somehow the ideal uh place that I want or paper to be used that uh it makes a framework uh that we can think in a very formal way that if I wanted to design an experiment to distinguish these two different Notions what would my generated what was the necessary necessary features that my generative model need to have and how can I uh make the realization of that generative model in an experimental paradigm what we do for example in the preprint uh is not in the main paper using a preprint that is online it's a and bio archive with the same set of authors there we try to find situations that for example it's a some election situation that each week before the election is on the news media tell you that uh this party is going to win with this probability then you see the results and we try to make some assumption that what are the features of these uh experiments and what we see is that oh actually different definitions make very different predictions so I cannot give a systematic approach that oh okay this is the algorithm that if you want to design an experiment just go one by one these steps but I want to say that this framework gives us the tools to think about before doing the experiment how to design it to have the maximal effect in many of these also indistinguishability conditions I say for example that if something is flat these two definitions are indistinguishable the thing is that if that distribution is not flat the two are distinguishable but the effects that is very small if it is close to flat so the best thing is that if I want to make two uh definition of surprise is indistinguishable in an experiment then it's best to look at that condition and make the opposite of that condition in my experiment and in that way I make the FX size as big as possible in some sense it's a very um non-rigorous way of talking about uh the theory but just intuitively that's how we design those experiments and could convince or experimental colleague at the moment to run the experiment so we are doing that awesome yeah there's a lot of ways that we could imagine pre-registration of studies and just the development of templates and patterns for statistical analyzes so that different kinds of surprise could be highlighted and we could say if the change point is going to be like this then you will be able to distinguish with this power these kinds of outcomes versus another um Ali do you want to question or yeah I actually this discussion reminds me of uh Terence Deacon's uh famous paper uh what is missing from theories of information uh because uh in that paper he also claims that one of the

misunderstandings about uh all the uh theories of information or all the relevant uh theories for that matter is the assumption that in order for any information to have a kind of for any content of information to have any real world consequences it must have some substantial properties and must correspond to some real world situations or something present as an extrinsic entity in some form another but as he shows in that paper we can regard the information whether as a kind of probability distribution or any kind of other theoretical construct as exactly conveying the kind of information that uh we want we wanted to explore in a much real sense and it's not just pure data or pure uh abstract information for that so for instance uh Shannon's analysis of information of course is based on the constraint and the entropy but on the other hand the capacity to convey that information is uh it depends on the relation to something that is specifically absent or not produced so this absence or presence of the entities that the information is about that entity or so-called the aboutness of the information is I heard the aboutness of the information but we'll pause and see if he'll rejoin otherwise we can continue I'll just kind of do you want to provide a thought on Shannon or just uh yeah that's actually very um important point that uh Ali was about to talk about uh which is in all these uh definition that I propose we didn't really care about uh what is uh why or what is X we were working in these very abstract States and a very abstract framework one may say that what if I don't care about these observations at all and I still may have some predictions and uh so this aboutness that you were talking about is quite important and whether I no matter how confident I am or what is my prediction how much I care about that problem is quite important and how surprised I am if I don't care about it at all then I may not feel surprised at all um so that's something that's really missing from the whole framework that how we can model the the the other things that have let's say semantic because at the moment uh all the framework is semantic free uh how how can we include some uh semantic into this framework we can talk about uh then we have another dimension to talk about surprise yes and some hope that with Progressive syntax schemes we can partition that semantics into other surprise definitions but the bigger question is what are we surprised about and I think that connects to these other cognitive phenomena slash complex word complexes like attention confidence salience um and I mean just to kind of put a point on that like when surprise is being used as an imperative for optimization and inference or merely as a signal that plays into action selection your robot or Aunt or person or software is gonna do different things depending on the surprise formalisms used and knowing many flavors of formalism it would be awesome if we could have a software package that computes 18 different numbers but then

what would we do we could choose one we could blend them we could make summary statistics we could say that we're in zone a where these five are aligned but this one isn't and then what would we do with that action how surprised would we be by that location and phase space so it's like a very interesting top-down situation independent phrasing of those relationships but in the implementation it's always going to come down to what the modeler chooses there just isn't a general answer to be found in the specifics one other thought or area to explore is you mentioned the Bayes factor and its role in learning so could you speak to optimal Bayesian inference and optimal what do you mean sorry and you like Bayes optimal inference okay that's what I meant that base Factor surprise appear in base optimal learning and also appear in the variational uh Bayesian update Rule and that was actually what we found cool about it because neuroscientists particularly people talking about the synaptic plasticity talked about surprise and how neuromodulators influence synaptic plasticity and we saw that if you're looking at the Bayesian framework we can find that such normative modulation based on surprise exist in how the belief is updated and it is funny that the same thing appear in lots of approximate inference scheme as well and it seems necessary to use such a surprise measures to modulate learning as we learn and there are lots of experimental evidence for it um so what I want to say is not necessarily only about the exact base and inference but it is something that appeared there and is useful for also other kind of uh other approximation to exact inference so what does optimal learning in this case mean optimal learning means to find the exact posterior given the model for those parameters because if the model is true if I believe that this is the true model the optic all the information that I can have about the hidden variable is in the posterior distribution of that variable given the observations and basically what we mean by learning is to um updating this uh posterior distribution from one time to another so as we go on and add more and more data points how we track this distribution over time and how we update our belief and by Optimal learning we mean the exact up Bayesian update from time T to $T + 1$. but this could be done also with variational inference but the cool thing was that also Bank variational inference would benefit from such modulation so one thought on that in a little bit of a meta science area is Bayesian statistics is becoming more prevalent and one way of reporting the effect of a treatment would be like with a base factor to present the relative evidence for one model versus another and something that that plays a functional similarity with but of course a technical difference is a p-value and so in the presentation I thought about how the base Factor was associated with learning as well as representing the evidence for two different possibilities in a ratio whereas a p-value

especially when we put um a discontinuity 0.05 is like a change point it's like we're sampling from the experimental outcomes and eggs are healthy eggs are healthy eggs are healthy now there's a difference between the groups so that puts the task of Science and learning into a change Point detection where oh now we're in the world where this is the case versus the sort of unrolling Bayesian approach where we get some experimental results that are all over the posterior but merely drawing results from a potentially quite dispersed posterior doesn't mean that we've entered any change point with respect to the efficacy of a given treatment I'm trying to uh to connect what you said about science in general to what you were thinking just as a point that the base Factor surprise we really took the word base factor from the field of statistics for the nature of hypothesis testing that exists in these kind of change Point detection because at each point you want to see that oh okay was there really a change in the environment or not and uh for example for this uh weather forecast in the pandemic I really had this before knowing that uh this is really a phenomenon and weather forecasts get less accurate for a couple of weeks I was confused that okay what's going on why my uh metal Swiss is not working uh well anymore so it's always some sort of hypothesis testing that okay am I living in the same world that it was yesterday or something changed from yesterday so there is some hypothesis testing and in the Bayesian setting it's done by comparing two models to base Factor but we can also have even in the Bayesian setting if we go again towards approximate Bayesian setting and sampling like the particle filtering we sample this change pouring so at some point we make decisions that uh oh this is the probability of for example there's a change or not at some point I should decide I would say yeah there is a change and from today onward I behave as if something changed yesterday and uh in a frequentist approach is a bit difficult for me to include things like p-value in the framework that I was talking about because the nature of p-value is in talking about probability of things more extreme than things that already happened while in this Bayesian setting we are talking about this particular observation I don't care about the values higher than observation lower than observation different from I I care about only this observation well in p-value we compute the probability of observation more extreme than this one and I don't see how it can easily fit in the framework but if I want to find the closest measure to say that it's a bit similar to p-value I would say it's something like Shannon's surprise because it doesn't have the nature of comparison you just say that under the current belief which let's say that the null hypothe is the current hypothesis how unlikely is the new observation yes one could certainly apply a change Point base Factor over three means we publish the paper base Factor under three is no

evidence and there's definitely some subtlety there with what the P statistic and other empirically calculated statistics tell us I'll read one question from the chat then Ali so Dave writes have folks working on these problems modeled the situation where counter evidence has no impact on confidence delusion and cognitive dissonance are terms that come into that discussion I'm sorry can you read again because I opened the page to read it myself as well but then all good have folks working on these problems model the situation where counter evidence has no impact on confidence delusion and cognitive dissonance are terms that come into that discussion that counter evidence doesn't have effect and confidence that was yes potentially where um surprising beliefs are continually entertained how could we model a situation where persistence uh cognitive dissonances or delusive beliefs are stabilized about cognitive doesn't when I talk about hypothesis testing in a Bayesian setting the fun thing is that all hypotheses live at the same time you just change the weight over them and some of these hypothesis are in disagreement with each other so this counterfactual assumptions that oh what if there was a change a while ago or what if uh oh wait no the previous observations are fixed I have this uh I have this imagination only in the unobserved States and when I talk about uh different scenarios I talk about unobserved States what I think the question was about if I want to think about counterfactuals then I should imagine that how would the world be if the observation was different and that at the moment is not included in the framework but one could think about it if we would uh integrate the framework it's a bit of causal inference and in those settings we have this counterfactual and talking about now I don't only condition in the past I imagine that some of the observation is very different and now I do my inference but I'm not sure if I got the question and if my answer was relevant I hope it is uh related awesome and on that um response Ian wrote cognitive dissonance I think there's something to do with having different beliefs at different levels of consciousness so yes these taxonomies were presented at a single level model as presented and with nested models we might have um different interactions amongst surprise definitions maybe even different surprise definitions playing out within a nested Bayesian architecture maybe one type of surprise is more amenable to Edge detection in the retina and then in a different part of the generative model there's another Opera operationalization of surprise in a decision making or in a risk averse setting of course that's uh definitely the case that different level of processing may work with different levels of surprise and um maybe I don't know uh belief mismatched price definitions are not relevant for the primary uh part of Cortex at all but if that's the case yeah I understand but uh maybe I misunderstood what they mean by cognitive Ollie any more thoughts or

questions maybe we can each have one more question or contribution yeah actually uh first of all I apologize I got disconnected in the middle of my question I don't know uh how much it was fine I could yeah uh all right so one thing about these kind of nested hierarchies um is that in the interactivist framework uh they usually uh Define different kinds of information as different levels of emergence in a kind of nested hierarchy uh the biggest of all is the Shannon information which deals with the kind of medium capacity of um conveying the information and then uh on another emergence level is uh the referential information which is about the aboutness and um at the highest emergence level we have the significant information or uh as they would like to call the usefulness or the semantic content of the information as they say so my question is that uh this kind of tax taxonomic approach that you uh provided in this paper uh how can we map this taxonomic system or even whether we can map this kind of taxonomic system into this kind of nested hierarchies of different information because obviously uh we can probably even say that the nature of these different kinds of informations are different right um my short answer would be the table that I showed if we want to account for the things that you discussed we should make a few of those tables so we should add another dimension so I don't think it's really possible to map this particular definitions to those different uh categories and for the long answer I give again another short answer which is basically we need to make a new framework to really include all these semantics or current probabilistic framework the one that I presented today doesn't have the capacity to uh account for that yeah that's why I mentioned that in manifold thing I said my first question because I was thinking of something similar because uh maybe uh in order to generalize this framework we would need a lot more Dimensions than uh just the two Dimensions that we have already provided in the paper so uh I don't know if as you say if it would be useful for any uh future research or whether it would be any kind of attractable uh computational framework but at least in my opinion or from purely theoretical point of view maybe it's worth investigating to see how these kinds of different Frameworks uh integrate into a much more generalized way of thinking about information and surprise but as I said your paper is a very important and significant contribution in that regard so in near closing the taxonomy notion it's a classification scheme and just as in Biology early Natural History observations of different things led to taxonomics and the classification of different things and on through phylogenetics and understanding the the temporal relationships so today we have a taxonomy for surprise that we didn't have a year ago what can be contributed by your group and by others in this emerging ecology of formalisms if we have the taxonomy today just a sketch

napkin sketch two by two grid we sorted the birds that people saw into roughly these categories they don't even know that it was this bird but now we can disambiguate it was that bird where will you be heading in this space and where do you think there are areas for people to contribute to this living area that's a very good point a very good question uh there is this slide that I had these different four domains of uh learning exploration memory and physiological signals that people talk about surprise in neuroscience and psychology in the middle these different definitions I think if you want to go beyond the taxonomy we should try to do more computational modeling and see that which of these definitions which part of these two degrees is related to which of these uh function so basically now I'm just proposing some tools and a framework which can be used later on for computational modeling so all these definitions can be useful or not for some modeler to explain exploration in some task or learning in some other tasks or segmentation in some task so going to be the next step is really finding out where and what function this definition makes sense yes you brought up the novelty curiosity learning and surprise is woven sometimes technically or not into those discussions but it's not all definitions a surprise for all definitions of learning salience attention novelty so disambiguating that space is very important Ollie any final remarks uh I would like to thank you again for inviting me and thanks a lot for all the discussion and nice question oh awesome it's a great honor and learning experience for us so you're always welcome back till next time you're really appreciate it thanks any reason thank you have a nice day bye-bye very well bye